

Live Acceptance Technical Brief

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INTERNAL

EXECUTIVE SUMMARY

Aegis proposes the 'Sovereign On-Premise Agentic AI Workbench' utilizing open-weight multimodal LLMs for confidential industrial work. The solution prioritizes 100% on-premise processing, intelligent model routing, agentic task execution, multimodal document understanding (PDFs, images, drawings), local RAG, and secure sandbox environments. Key differentiators include data sovereignty, adaptability through modular models, and a dynamic model loading system.

KEY FINDINGS

The Sovereign AI Workbench focuses on secure, local AI execution, rejecting cloud API reliance. The core architecture involves an 'Agent' that plans and executes tasks, utilizing a 'Model Registry' with quantized 7B/8B LLMs (text & code) and a vision/OCR model. Dynamic model loading (1. Task Arrives, 2. Check Required Model, 3. Load If Needed, 4. Execute, 5. Unload If Idle) optimizes resource utilization. The system supports a wide range of multimodal inputs (text, PDFs, scans, images, drawings) and outputs (DOCX, XLSX, PPTX, PDF, code). Operational components include a secure web UI, an agent controller, a document workflow (PDF summarization, RAG), a coding workflow (code generation, execution), and a multimodal workflow (visual analysis). The system incorporates robust security measures like audit logs, local network isolation, and user roles.

DETAILED ANALYSIS

The system's architecture is modular, allowing for easy addition or modification of AI models. The 'Agentic' workflow employs a plan-execute-verify-re-plan iterative process. Model selection is driven by task requirements and utilizes a dynamic loading mechanism to minimize latency. The system incorporates RAG (Retrieval-Augmented Generation) over organizational knowledge for enhanced accuracy and context. The sandbox environment enables safe code execution and reduces the risk of data leakage or malicious code. Quantization of models is implemented to reduce memory footprint and improve performance, particularly on GPUs. The system's multimodality supports diverse industrial workflows, including those involving scanned documents and visual data. The data remains entirely within the organization's network, ensuring data sovereignty and control.

RISKS AND OPERATIONAL ISSUES

Potential risks include GPU limits and contention, requiring careful resource management. The OCR + VLM + Document pipeline introduces complexity and potential for hallucination. The system's reliance on local processing necessitates robust network isolation to prevent unauthorized access. Validation of generated outputs (accuracy, safety, consistency) is a critical operational requirement. The 'Re-plan' feature introduces potential for complexity and increased processing time. Managing the model registry and dynamic loading adds operational overhead. The use of quantized models might require trade-offs between performance and model

size.

RECOMMENDATIONS

Prioritize robust GPU resource management and monitoring to mitigate contention. Implement a comprehensive validation process for generated outputs to ensure accuracy and safety. Invest in a detailed audit trail and logging system to track agent actions and user interactions. The system's modular architecture should be leveraged to enable iterative model updates and improvements. Explore optimization techniques for the OCR and vision components to minimize hallucination. Further research into the effectiveness and stability of dynamic model loading is recommended.

AUTHORITATIVE SOURCES & CITATIONS

Document Name	Referenced Pages	Relevance Status
SIH2026ppt.pdf	Page(s) 1, 2, 3, 4, 5, 6, 7	High

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